

Practice Exercise - Repetition structure

For-Loops

1. Counting Sheep

A shepherd is counting sheep before sleeping. Each time a sheep jumps over the fence, the shepherd counts it. The counting stops after 10 sheep.

Write a for loop that prints numbers from 1 to 10, representing the sheep jumping over the fence.

2. Rocket Launch Countdown

A rocket is preparing for launch. The countdown starts from 10 and ends at 1 before liftoff.

Write a for loop that counts down from 10 to 1 and then prints "Liftoff!"

3. Module Average

Students want to calculate their university mark by providing all their module marks. They want to use this to see whether they can apply for honours (minimum 60% average)

Write a program that asks the user how many modules they want to enter and then the students will have to enter each module mark to calculate an average.

4. Factorial Calculation

A scientist is calculating the factorial of a number to solve a probability problem.

Write a program that calculates the factorial of a number input by the user.

Eg, $5! = 5 * 4 * 3 * 2 * 1 = 120$

5. Fibonacci Sequence (Self-study)

The Fibonacci sequence starts with 0 and 1, and each new term is the sum of the previous two.

Write a for loop that generates the first n terms of the Fibonacci sequence, where n is input by the user.

Eg, first 7 terms will be: 0 1 1 2 3 5 8

While-Loops

6. Password checker

A user wants to gain access to their super-secret program. The system has a built-in password which is "P@55w0rd123!".

Write a program that continuously checks whether the password is correct or not, if the password is incorrect, print "Access DENIED! Please try again"

7. Entering a Positive Number

A user is asked to enter a positive number. If they enter a negative number or zero, they are prompted again until they enter a valid number.

Write a while loop that keeps asking for a positive number until the user enters one.

8. Climbing the Mountain

A climber is trying to reach the peak of a 5000m mountain. Each day, they climb can only climb between 300m and 700m but slide down 150m at night. The user should enter how much they climbed each day.

Write a program that simulates the climb and prints how many days it takes to reach the top. Be sure to tell the climber how much they still need to climb after the slid down at night.

9. Guess the Number (Unlimited Tries but keep count!)

A computer randomly selects a number between 1 and 100, and the user must guess it. The program gives hints: "Too high" or "Too low."

Write a while loop that continues until the user guesses the correct number.

HINT:

- Makes use of the random module (`random.randint(1, 100)`)

10. Detect Prime Numbers (Self-study)

A number is considered prime if it has no divisors other than 1 and itself.

Write a program that asks the user for a number and checks if it is prime. The program should keep asking until the user enters a prime number.